

Crop Disease Diagnosis Expert(CROPDX) System

Prolog Project

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Submitted to:

Dr. Mohammad Shahadat Hossain

Professor

Department of Computer Science and Engineering

University of Chittagong

Submitted by:

Humayra Adiba

ID: 22701050

Department of Computer Science & Engineering

University of Chittagong

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Special thanks go to our classmates for healthy academic discussions, and to the open-source Prolog community whose archived resources on Turbo Prolog proved indispensable.

This project, however modest in scope, has deepened our appreciation for rule-based artificial intelligence — a paradigm that continues to underpin modern decision-support systems in agriculture, medicine, and beyond.

Humayra Adiba

ID: 22701050

Department of Computer Science and Engineering

University of Chittagong

April 30, 2026

Abstract

Abstract

Crop diseases are a leading cause of agricultural yield loss worldwide, particularly in developing economies where access to specialised agricultural expertise is limited. This project presents a **rule-based expert system** — *Crop Disease Diagnosis System (CROPDX)* — implemented in **Turbo Prolog** that simulates the diagnostic reasoning of an experienced agronomist.

The system supports four staple crops — **Rice, Potato, Tomato, and Wheat** — and is capable of identifying eight distinct diseases through a structured yes/no symptom questionnaire. For each identified disease the system outputs (i) the confirmed diagnosis, (ii) the causative organism, (iii) a recommended treatment protocol, and (iv) preventive agronomic practices.

The knowledge base encodes pattern-matching rules derived from standard phytopathology literature. Prolog's inherent backward-chaining inference engine performs the pattern matching, providing a transparent and auditable reasoning process suitable for educational and advisory purposes.

Implemented on DOSBox under Windows, the system demonstrates that classical Artificial Intelligence(AI) paradigms remain relevant for low-resource, offline deployment in rural agricultural settings. The project achieved all stated objectives: multi-crop coverage, dual-disease differentiation per crop, treatment advice, and graceful handling of unrecognised symptom patterns.

Keywords: *Expert System, Turbo Prolog, Rule-Based Reasoning, Crop Disease Diagnosis, Knowledge Base, Agricultural AI, Pattern Matching.*

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1 Introduction

Agriculture forms the backbone of many developing economies, providing food security and livelihoods for billions of people. However, crop diseases pose a persistent and severe threat to agricultural productivity. Early and accurate diagnosis is critical — delayed or incorrect diagnosis can result in the total loss of a season’s harvest.

In many rural areas, access to trained plant pathologists is scarce. Farmers must often rely on intuition or seek advice from distant experts, leading to costly delays. **Expert systems** — computer programs that encode expert knowledge in a form accessible to non-experts — offer a compelling solution to this problem[4].

What is an Expert System?

An **expert system** is an AI program that mimics the decision-making ability of a human expert within a specific domain. It consists of:

- A **Knowledge Base** — facts and rules about the domain.
- An **Inference Engine** — the reasoning mechanism.
- A **User Interface** — for interaction with the end user.

This project implements a crop disease diagnosis expert system using **Turbo Prolog**, a logic programming language based on first-order predicate logic. The system guides a farmer (or agricultural extension worker) through a structured symptom questionnaire and returns a diagnosis along with actionable advice.

Turbo Prolog was selected because its built-in backward-chaining inference engine aligns naturally with the rule-based architecture of expert systems, and because it provides an excellent pedagogical context for understanding AI fundamentals[1].

1.1 Scope of the System

The current version of CROPDX covers:

- **4 crops:** Rice, Potato, Tomato, Wheat
- **8 diseases:** 2 per crop
- **24 diagnostic rules** encoded in Prolog clauses
- **Interactive CLI** with yes/no symptom elicitation

2 Problem Statement

Core Problem

Farmers in developing regions lack timely access to agricultural experts capable of diagnosing crop diseases. Without prompt, accurate diagnosis, incorrect or excessive pesticide application leads to financial loss, environmental harm, and continued crop damage.

Specifically, the following pain points motivated this project:

1. **Geographical isolation:** Agricultural experts are concentrated in urban centres; rural farmers have minimal access.
2. **Symptom misidentification:** Many diseases produce visually similar symptoms (e.g. Rice Blast vs. Bacterial Leaf Blight both cause leaf discolouration), leading to misdiagnosis.
3. **Delayed treatment:** Waiting for expert consultation allows disease to spread, compounding losses.
4. **Lack of digital tools:** Internet connectivity in rural Bangladesh and similar regions remains unreliable, making cloud-based solutions impractical.
5. **Pesticide overuse:** Without accurate diagnosis, farmers may apply broad-spectrum pesticides unnecessarily, increasing costs and pesticide resistance.

An offline, rule-based expert system running on minimal hardware addresses all five pain points simultaneously.

3 Aims and Objectives

3.1 Primary Aim

To design and implement a functional expert system in Turbo Prolog that can diagnose common diseases of four staple crops through symptom-based rule matching, and provide actionable treatment and prevention advice.

3.2 Specific Objectives

1. Encode domain knowledge (symptoms, diagnoses, treatments) for 4 crops and 8 diseases into a structured Prolog knowledge base.
2. Implement a user-friendly interactive interface using `write/readln` predicates.
3. Design diagnostic rules that correctly differentiate diseases with similar symptom profiles.
4. Provide a graceful fallback when no disease pattern is matched.
5. Implement session management allowing multiple diagnoses per run.
6. Ensure full compatibility with Turbo Prolog syntax and DOSBox deployment.
7. Produce structured output (Disease $\rightarrow$ Cause $\rightarrow$ Treatment $\rightarrow$ Prevention).

4 Literature Review

4.1 Expert Systems in Agriculture

Expert systems have been applied to agricultural diagnosis since the 1980s. The pioneering PLANT/ds system [11] diagnosed soybean diseases using 800+ rules and demonstrated the feasibility of AI-assisted agronomy. Subsequently, CALEX/Wheat and COMAX extended these ideas to pest management and cotton crop scheduling.

4.2 Prolog and Logic-Based AI

Prolog (Programming in Logic) was created by Alain Colmerauer in 1972 and remains one of the most natural languages for knowledge representation[2]. **Turbo Prolog**, released by Borland International in 1986, introduced strong typing, a compiled execution model,

and an integrated development environment, making Prolog accessible to a broader developer community[3].

Key advantages of Prolog for expert systems:

- Built-in backtracking matches rule-based reasoning naturally.
- Clauses read declaratively — they state *what* is true, not *how* to compute it.
- Easy to extend the knowledge base without modifying the inference engine.

4.3 Crop Disease Identification Research

Disease		Pathogen[6]	Economic Impact[8]
Rice Blast		<i>Magnaporthe oryzae</i>	Up to 30% yield loss annually worldwide[5]
Bacterial Leaf Blight		<i>Xanthomonas oryzae</i>	Major constraint in Asian paddy systems
Late Blight		<i>Phytophthora infestans</i>	Caused Irish Famine (1845); still devastating
Early Blight		<i>Alternaria solani</i>	20–30% yield reduction in un-managed fields
Tomato Mosaic Virus		TMV	Spreads rapidly; no chemical cure
Fusarium Wilt		<i>F. oxysporum</i>	Soil-borne; persists for years
Yellow Rust		<i>Puccinia striiformis</i>	Global wheat losses \$1B+ annually
Powdery Mildew		<i>Blumeria graminis</i>	Reduces grain fill; quality loss

5 Methodology

5.1 Development Approach

The project followed a **Knowledge Engineering** methodology consisting of five phases:

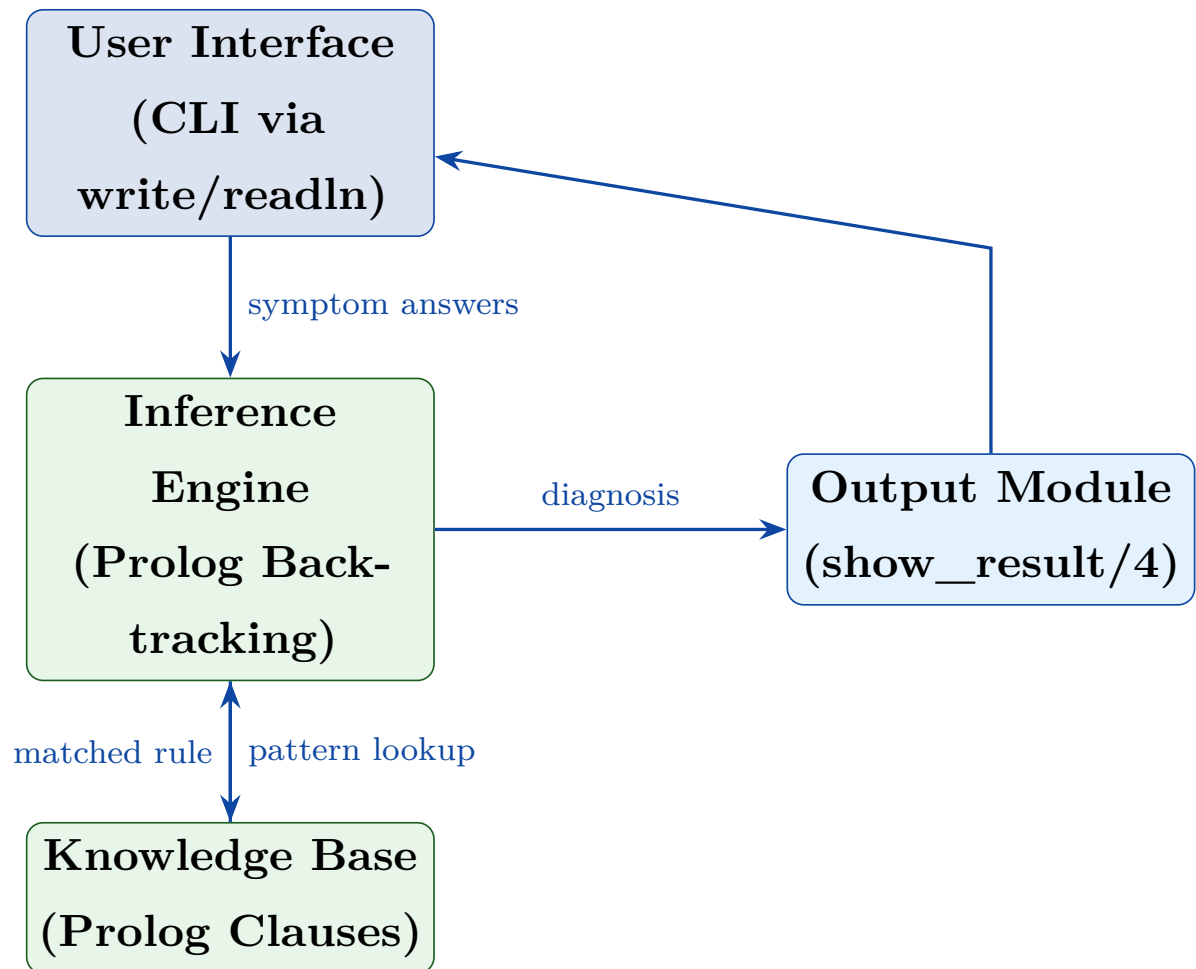
1. **Domain Analysis** — Study of phytopathology references and identification of the 8 most diagnostically distinct diseases.
2. **Knowledge Acquisition** — Extraction of diagnostic rules: symptom sets, causative organisms, and treatment protocols.
3. **Knowledge Representation** — Encoding rules in Turbo Prolog's clause format.
4. **System Implementation** — Writing, compiling, and testing `CROPDX.PRO` in the DOSBox/Turbo Prolog environment.
5. **Validation** — Tracing the inference engine through each diagnostic path to verify correctness.

5.2 Tools and Environment

Component	Details
Programming Language	Turbo Prolog
Emulation Platform	DOSBox 0.74 on Windows 10
IDE	Turbo Prolog Integrated Editor
Compilation	Alt+C (internal compiler)
Execution	F9 (compile & run)
Source File	CROPDX.PRO

6 System Design

6.1 High-Level Architecture



6.2 Input Process

The user interaction is handled by the `ask_question/2` predicate, which prints each symptom question and reads the user's yes/no response via `readln/1`. Six questions are asked per crop, stored as answer variables A1–A6.

6.3 Symptom Analysis

The six symptom answers are passed to the crop-specific diagnosis predicate (e.g. `diagnose_rice/6`). Turbo Prolog's top-down clause matching evaluates each diagnostic rule in sequence.

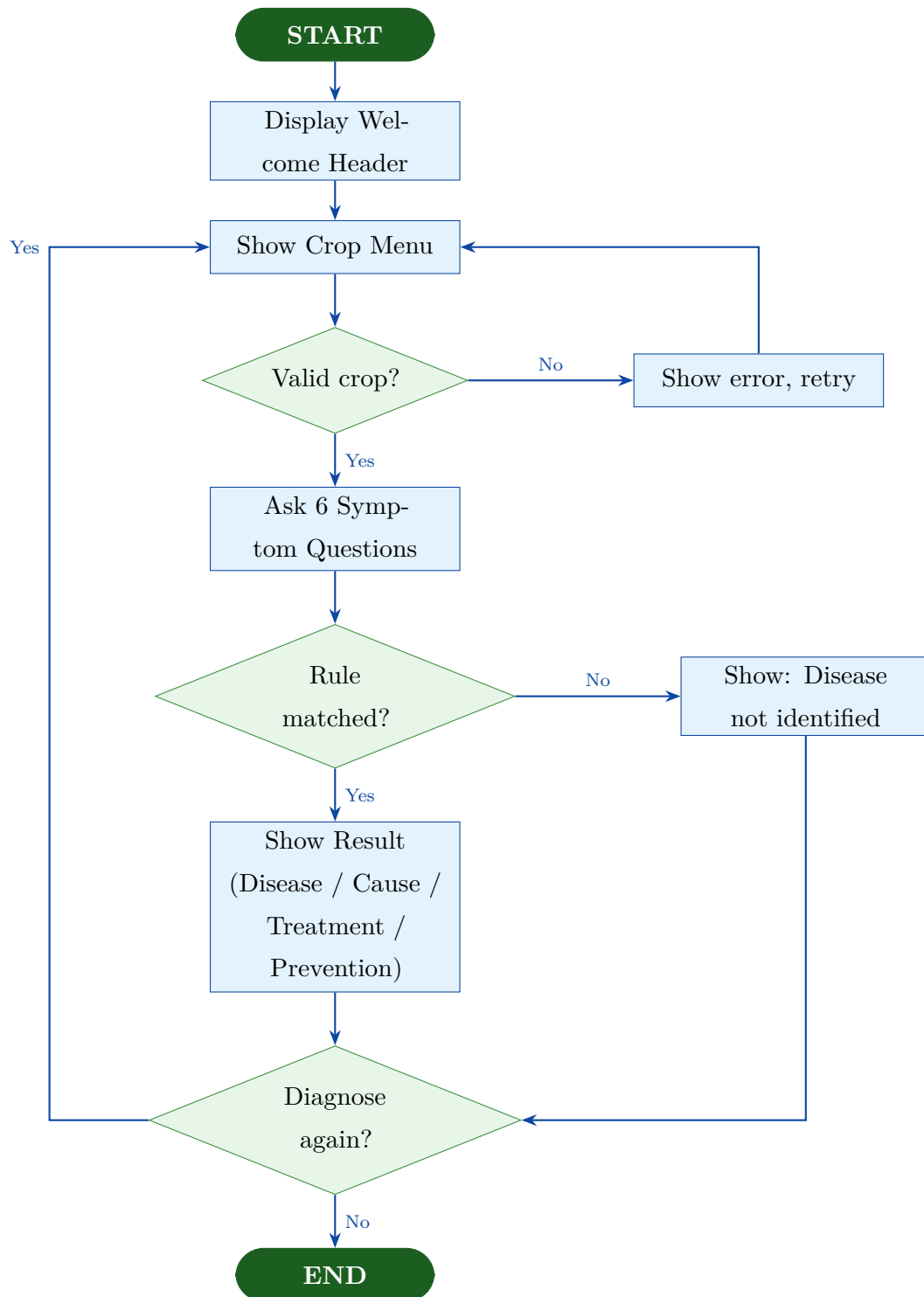
6.4 Disease Diagnosis

Each disease is represented by a clause head whose argument pattern encodes the required symptom set. A pattern of `"yes","yes","yes",_ ,_ ,_` matches Disease 1; `_ ,_ ,_ ,"yes","yes","yes"` matches Disease 2. The underscore (`_`) acts as a wildcard.

6.5 Treatment Recommendation

Upon a successful match, `show_result/4` is called with four string arguments: Disease name, Cause, Treatment, and Prevention. The output is formatted for clarity using separator lines.

6.6 System Flowchart



7 Algorithm / Working Procedure

The system's reasoning algorithm is described below in structured pseudocode derived directly from the Prolog implementation.

Algorithm: Crop Disease Diagnostic Engine

```
PROCEDURE CROPDX_Diagnose():
    CALL show_header()
    CALL main_menu()

PROCEDURE main_menu():
    PRINT crop options (rice/potato/tomato/wheat/exit)
    READ CropName
    CALL select_crop(CropName)

PROCEDURE select_crop(CropName):
    MATCH CropName:
        "rice"    -> CALL ask_rice(A1..A6); CALL diagnose_rice(A1..A6)
        "potato"  -> CALL ask_potato(B1..B6); CALL diagnose_potato(B1..B6)
        "tomato"  -> CALL ask_tomato(C1..C6); CALL diagnose_tomato(C1..C6)
        "wheat"   -> CALL ask_wheat(D1..D6); CALL diagnose_wheat(D1..D6)
        "exit"    -> PRINT goodbye; TERMINATE
        _         -> PRINT error; CALL main_menu()

PROCEDURE ask_[crop](V1..V6):
    FOR each symptom question Q[i]:
        PRINT Q[i]; READ V[i]

PROCEDURE diagnose_[crop](V1..V6):
    IF V1="yes" AND V2="yes" AND V3="yes":
        CALL show_result(Disease1, Cause1, Treatment1, Prevention1)
    ELSE IF V4="yes" AND V5="yes" AND V6="yes":
        CALL show_result(Disease2, Cause2, Treatment2, Prevention2)
    ELSE:
        CALL show_unknown()
```

```
CALL ask_again()
```

The Prolog inference engine evaluates clause heads top-to-bottom (**top-down resolution**) and backtracks automatically if a head fails to unify with the goal[1]. The `_` wildcard in clause heads enables partial pattern matching, a cornerstone of expert-system rule design.

8 Knowledge Base Design

8.1 Knowledge Representation Model

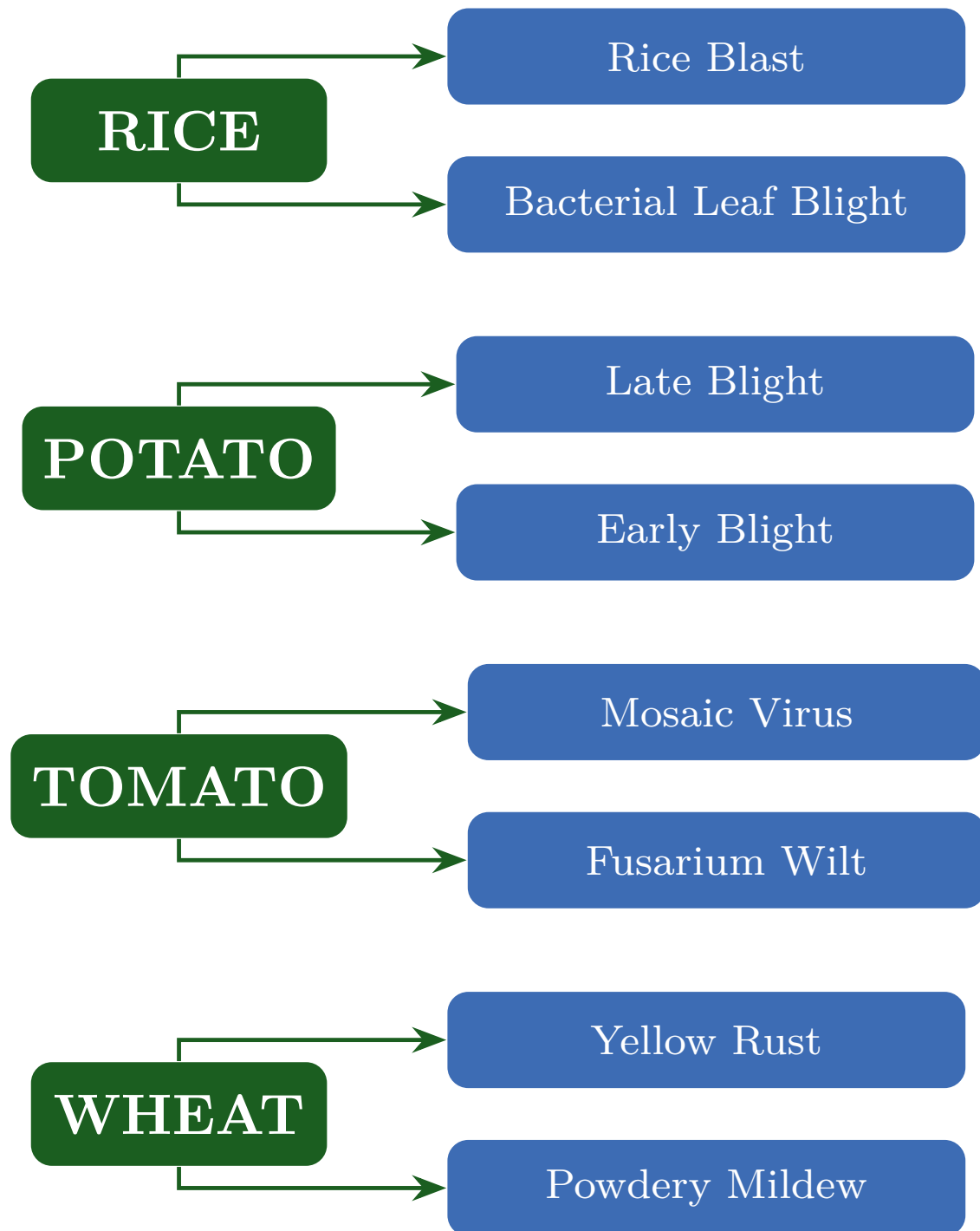
The knowledge base is represented as a set of Prolog clauses encoding the **symptom** $\rightarrow$ **disease** mapping for each crop:

$$\text{diagnose_crop}(S_1, S_2, \dots, S_6) \leftarrow S_i \in \{\text{“yes”,_}\}$$

8.2 Disease–Symptom Mapping

Crop	Disease	Diagnostic Symptoms	Pathogen
Rice	Rice Blast	Diamond spots; gray/white centre; brown border	<i>M. oryzae</i>
Rice	Bacterial Leaf Blight	Yellow-wilting tips; water-soaked streaks; tip drying	<i>X. oryzae</i>
Potato	Late Blight	Dark water-soaked patches; white under-leaf mold; tuber rot	<i>P. infestans</i>
Potato	Early Blight	Small brown spots; yellow halo; lower leaves first	<i>A. solani</i>
Tomato	Mosaic Virus	Mosaic pattern; leaf curl; stunted growth	TMV
Tomato	Fusarium Wilt	Lower wilting; night recovery; brown stem	<i>F. oxysporum</i>
Wheat	Yellow Rust	Yellow pustules; pustule rows; pale yellowing	<i>P. striiformis</i>
Wheat	Powdery Mildew	White powdery patches; rubs off; upper leaf	<i>B. graminis</i>

8.3 Crop–Disease Mapping Diagram



9 Turbo Prolog Implementation

9.1 Program Structure Overview

The source file `CROPDX.PRO` is organised into four mandatory Turbo Prolog sections:

Section	Purpose	Example
<code>domains</code>	Type declarations	<code>name = string</code>
<code>predicates</code>	Predicate signatures	<code>diagnose_rice(answer, ...)</code>
<code>clauses</code>	Rules and facts	<code>diagnose_rice("yes",...) :- ...</code>
<code>goal</code>	Entry point	<code>start.</code>

9.2 Key Predicate Roles

9.2.1 `start/0` and `show_header/0`

Entry predicates that print the application banner and chain into the main menu via `main_menu/0`.

9.2.2 `main_menu/0` and `select_crop/1`

The navigation layer. `readln/1` captures the crop name as a string; pattern-matching clauses route execution to the appropriate crop module. The final wildcard clause handles invalid input with a re-prompt loop.

9.2.3 `ask_[crop]/6` Predicates

Each crop has a dedicated six-question symptom questionnaire predicate. Questions are posed via `ask_question/2`, which combines `write/1` and `readln/1`. The six answers are bound to variables and passed directly to the corresponding `diagnose_[crop]/6` predicate.

9.2.4 `diagnose__[crop]/6` Predicates

The core diagnostic engine. Each crop has three clauses: Disease 1 pattern (first three arguments "yes"), Disease 2 pattern (last three arguments "yes"), and a wildcard catch-all. Turbo Prolog evaluates clauses in declaration order — the first clause whose head unifies with the symptom pattern succeeds and calls `show_result/4`.

The **cut operator** (!) is placed after each successful `show_result/4` call. This is critical: without !, Prolog backtracks after printing the result and incorrectly executes the wildcard clause, producing a duplicate `show_unknown` output. The cut prevents this by committing to the first matching clause.

9.2.5 `show_result/4` and `show_unknown/0`

Output formatting predicates. `show_result/4` accepts four string arguments — disease name, cause, treatment, and prevention — and formats the complete diagnosis report. `show_unknown/0` provides a graceful fallback message when no symptom pattern matches, advising the user to consult an agricultural expert.

9.2.6 `ask_again/0` and `check_again/1`

Session management predicates. After each diagnosis, `ask_again/0` prompts the user to continue or exit. `check_again/1` uses two clauses: one matching "yes" to restart `main_menu/0`, and a wildcard clause to print the goodbye message and terminate.

10 Source Code

10.1 CROPDX.PRO — Full Annotated Source

Listing 1: Domains Section

```

1  /* =====
2     CROP DISEASE DIAGNOSIS EXPERT SYSTEM
3     File       : CROPDX.PRO
4     Language  : Turbo Prolog
5     Student   : Humayra Adiba | ID: 22701050
6     Course    : CSE 714 - AI Lab
7     =====
8
9     */
10 domains
11     name      = string    /* for disease names, crop names, questions
12                             */
13     answer    = string    /* stores "yes" or "no" from user input
14                             */

```

Listing 2: Predicates Section

```

1 predicates

```

```

2   start
3   main_menu
4   select_crop(name)
5
6   ask_rice(answer, answer, answer, answer, answer, answer)
7   ask_potato(answer, answer, answer, answer, answer, answer)
8   ask_tomato(answer, answer, answer, answer, answer, answer)
9   ask_wheat(answer, answer, answer, answer, answer, answer)
10
11  diagnose_rice(answer, answer, answer, answer, answer, answer)
12  diagnose_potato(answer, answer, answer, answer, answer, answer)
13  diagnose_tomato(answer, answer, answer, answer, answer, answer)
14  diagnose_wheat(answer, answer, answer, answer, answer, answer)
15
16  get_answer(answer)
17  ask_question(name, answer)
18  check_again(answer)
19  show_result(name, name, name, name)
20  show_unknown
21  show_line
22  show_header
23  ask_again

```

Listing 3: Entry Point and Header

```

1  clauses
2
3  start :-
4      show_header,
5      main_menu.
6
7  show_header :-
8      nl,
9      write("===== "
10         ),
11      nl,
12      write("    CROP DISEASE DIAGNOSIS EXPERT SYSTEM
13         "),
14      nl,
15      write("    AI Lab Project   |   Turbo Prolog 2.0
16         "),
17      nl,
18      write("    Humayra Adiba   |   ID: 22701050
19         "),

```

```
16     nl,
17     write("=====
18         "),
19     nl, nl.
20
21 show_line :-
22     write("-----"),
23     ,
24     nl.
```

Listing 4: Main Menu and Crop Routing

```
1  main_menu :-
2      nl,
3      write("Select your crop: "), nl,
4      write("  rice / potato / tomato / wheat / exit"), nl, nl,
5      write("Enter crop name: "),
6      readln(CropName), nl,
7      select_crop(CropName).
8
9  select_crop("rice") :-
10     show_line,
11     write("Crop selected: RICE"), nl, show_line,
12     ask_rice(A1, A2, A3, A4, A5, A6),
13     diagnose_rice(A1, A2, A3, A4, A5, A6).
14
15  select_crop("potato") :-
16     show_line,
17     write("Crop selected: POTATO"), nl, show_line,
18     ask_potato(B1, B2, B3, B4, B5, B6),
19     diagnose_potato(B1, B2, B3, B4, B5, B6).
20
21  select_crop("tomato") :-
22     show_line,
23     write("Crop selected: TOMATO"), nl, show_line,
24     ask_tomato(C1, C2, C3, C4, C5, C6),
25     diagnose_tomato(C1, C2, C3, C4, C5, C6).
26
27  select_crop("wheat") :-
28     show_line,
29     write("Crop selected: WHEAT"), nl, show_line,
30     ask_wheat(D1, D2, D3, D4, D5, D6),
31     diagnose_wheat(D1, D2, D3, D4, D5, D6).
32
```

```

33 select_crop("exit") :-
34     nl,
35     write("Goodbye! Thank you for using the system."), nl.
36
37 select_crop(_) :-
38     nl,
39     write("Invalid input. Please type: rice, potato, tomato,"),
40     nl,
41     write("wheat, or exit."),
42     nl,
43     main_menu.

```

Listing 5: Question Helper Predicates

```

1 get_answer(Answer) :-
2     readln(Answer), nl.
3
4 ask_question(Question, Answer) :-
5     write("    >> "), write(Question), nl,
6     write("        (yes/no): "),
7     get_answer(Answer).

```

Listing 6: Rice Symptom Questions

```

1 ask_rice(A1, A2, A3, A4, A5, A6) :-
2     nl,
3     write("Answer each question with yes or no:"), nl, nl,
4     ask_question(
5         "Do you see diamond-shaped spots on leaves?", A1),
6     ask_question(
7         "Are the spots gray or white in the center?", A2),
8     ask_question(
9         "Do the spots have brown or dark borders?", A3),
10    ask_question(
11        "Are the leaf tips turning yellow and wilting?", A4),
12    ask_question(
13        "Do you see water-soaked streaks on leaf margins?", A5),
14    ask_question(
15        "Are leaves drying out from the tips downward?", A6).

```

Listing 7: Rice Diagnosis Rules — with Cut Operator

```

1 /* KEY FIX: Cut operator (!) added after each rule.
2 Without !, Prolog continues searching after a match
3 and may print wrong results or show_unknown wrongly. */

```

```

4
5  /* Rule 1: Rice Blast -- first 3 symptoms all "yes" */
6  diagnose_rice("yes", "yes", "yes", _, _, _) :-
7      show_result(
8          "RICE BLAST",
9          "Caused by the fungus Magnaporthe oryzae.",
10         "Apply Tricyclazole or Carbendazim. Remove infected parts."
11         ,
12         "Use resistant varieties. Avoid excess nitrogen."
13     ), !.
14
15 /* Rule 2: Bacterial Leaf Blight -- last 3 symptoms all "yes" */
16 diagnose_rice(_, _, _, "yes", "yes", "yes") :-
17     show_result(
18         "BACTERIAL LEAF BLIGHT",
19         "Caused by Xanthomonas oryzae pv. oryzae.",
20         "Apply copper-based bactericides. Remove infected leaves.",
21         "Plant resistant varieties. Avoid waterlogging."
22     ), !.
23
24 /* Fallback: no pattern matched */
25 diagnose_rice(_, _, _, _, _, _) :-
26     show_unknown.

```

Listing 8: Potato Symptom Questions

```

1  ask_potato(B1, B2, B3, B4, B5, B6) :-
2      nl,
3      write("Answer each question with yes or no:"), nl, nl,
4      ask_question(
5          "Do you see dark brown water-soaked patches on leaves?", B1
6      ),
7      ask_question(
8          "Is there white mold visible under the leaves?", B2
9      ),
10     ask_question(
11         "Are tubers showing dark rotting areas inside?", B3
12     ),
13     ask_question(
14         "Do leaves show small brown spots with yellow rings?", B4
15     ),
16     ask_question(
17         "Are the spots surrounded by a yellow halo?", B5
18     ).

```

```

    ),
14   ask_question(
15       "Are older lower leaves affected first?",
                                           B6
    ).

```

Listing 9: Potato Diagnosis Rules

```

1  /* Rule 1: Late Blight -- first 3 symptoms all "yes" */
2  diagnose_potato("yes", "yes", "yes", _, _, _) :-
3      show_result(
4          "LATE BLIGHT",
5          "Caused by Phytophthora infestans.",
6          "Apply Mancozeb or Metalaxyl. Destroy infected tubers.",
7          "Use certified disease-free seed. Improve drainage."
8      ), !.
9
10 /* Rule 2: Early Blight -- last 3 symptoms all "yes" */
11 diagnose_potato(_, _, _, "yes", "yes", "yes") :-
12     show_result(
13         "EARLY BLIGHT",
14         "Caused by fungus Alternaria solani.",
15         "Apply Chlorothalonil or Mancozeb. Remove infected leaves."
16     ), !.
17
18
19 diagnose_potato(_, _, _, _, _, _) :-
20     show_unknown.

```

Listing 10: Tomato Symptom Questions

```

1  ask_tomato(C1, C2, C3, C4, C5, C6) :-
2      nl,
3      write("Answer each question with yes or no:"), nl, nl,
4      ask_question(
5          "Do leaves show a mosaic of light and dark green?",
6          C1),
7      ask_question(
8          "Are leaves curled, puckered, or distorted?",
9          C2),
10     ask_question(
11         "Is the plant stunted or growing slowly?",
12         C3),
13     ask_question(
14         "Are there any small holes or damage on the leaves?",
15         C4),
16     ask_question(
17         "Are there any small white spots on the leaves?",
18         C5),
19     ask_question(
20         "Are there any small dark spots on the leaves?",
21         C6).

```

```

11     "Are lower leaves yellowing and wilting during the day?",
        C4),
12     ask_question(
13     "Does wilting recover at night but worsen each day?",
        C5),
14     ask_question(
15     "Is the stem brown inside when you cut it open?",
        C6).

```

Listing 11: Tomato Diagnosis Rules

```

1  /* Rule 1: Tomato Mosaic Virus -- first 3 symptoms all "yes" */
2  diagnose_tomato("yes", "yes", "yes", _, _, _) :-
3      show_result(
4          "TOMATO MOSAIC VIRUS",
5          "Caused by Tomato Mosaic Virus spread by insects.",
6          "Remove infected plants. Control aphids with insecticide.",
7          "Use virus-resistant seeds. Sanitize tools."
8      ), !.
9
10 /* Rule 2: Fusarium Wilt -- last 3 symptoms all "yes" */
11 diagnose_tomato(_, _, _, "yes", "yes", "yes") :-
12     show_result(
13         "FUSARIUM WILT",
14         "Caused by soil fungus Fusarium oxysporum.",
15         "Remove infected plants. Apply Thiophanate-methyl.",
16         "Use Fusarium-resistant varieties. Rotate crops."
17     ), !.
18
19 diagnose_tomato(_, _, _, _, _, _) :-
20     show_unknown.

```

Listing 12: Wheat Symptom Questions

```

1  ask_wheat(D1, D2, D3, D4, D5, D6) :-
2      nl,
3      write("Answer each question with yes or no:"), nl, nl,
4      ask_question(
5          "Do you see yellow or orange stripe pustules on leaves?",
            D1),
6      ask_question(
7          "Are the pustules arranged in rows along the leaf?",
            D2),
8      ask_question(

```



```

9      "Do leaves turn pale yellow between the pustule strips?",
      D3),
10  ask_question(
11      "Do you see white powdery patches on leaves or stems?",
      D4),
12  ask_question(
13      "Are the powdery patches easily rubbed off?",
      D5),
14  ask_question(
15      "Are upper leaf surfaces primarily affected?",
      D6).

```

Listing 13: Wheat Diagnosis Rules

```

1  /* Rule 1: Yellow Rust -- first 3 symptoms all "yes" */
2  diagnose_wheat("yes", "yes", "yes", _, _, _) :-
3      show_result(
4          "YELLOW RUST (STRIPE RUST)",
5          "Caused by fungus Puccinia striiformis.",
6          "Apply Propiconazole or Tebuconazole at first sign.",
7          "Plant rust-resistant varieties. Monitor in cool wet
          weather."
8      ), !.
9
10 /* Rule 2: Powdery Mildew -- last 3 symptoms all "yes" */
11 diagnose_wheat(_, _, _, "yes", "yes", "yes") :-
12     show_result(
13         "POWDERY MILDEW",
14         "Caused by fungus Blumeria graminis.",
15         "Apply Triadimefon or Sulfur-based fungicide.",
16         "Use resistant varieties. Avoid dense planting."
17     ), !.
18
19 diagnose_wheat(_, _, _, _, _, _) :-
20     show_unknown.

```

Listing 14: Result Display and Session Management

```

1  show_result(Disease, Cause, Treatment, Prevention) :-
2      nl, show_line,
3      write("DIAGNOSIS RESULT"), nl,
4      show_line, nl,
5      write("DISEASE      : "), write(Disease),      nl, nl,
6      write("CAUSE         : "), write(Cause),          nl, nl,
7      write("TREATMENT    : "), write(Treatment), nl, nl,

```

```
8      write("PREVENTION: "), write(Prevention), nl, nl,
9      show_line,
10     ask_again.
11
12 show_unknown :-
13     nl, show_line,
14     write("DIAGNOSIS RESULT"), nl,
15     show_line,
16     write("Disease could not be identified."), nl,
17     write("Please consult a local agricultural expert."), nl,
18     show_line,
19     ask_again.
20
21 ask_again :-
22     nl,
23     write("Diagnose another crop? (yes/no): "),
24     readln(Again), nl,
25     check_again(Again).
26
27 check_again("yes") :-
28     main_menu.
29
30 check_again(_) :-
31     write("Thank you for using the system. Goodbye!"), nl.
32
33 /* =====
34    GOAL - Program Entry Point
35    =====
36    */
37 goal
38     start.
```

11 Sample Input and Output

11.1 System Screenshots

The following screenshots were captured during actual execution of CROPDX.PRO in the Turbo Prolog IDE running inside DOSBox 0.74-3 on Windows.

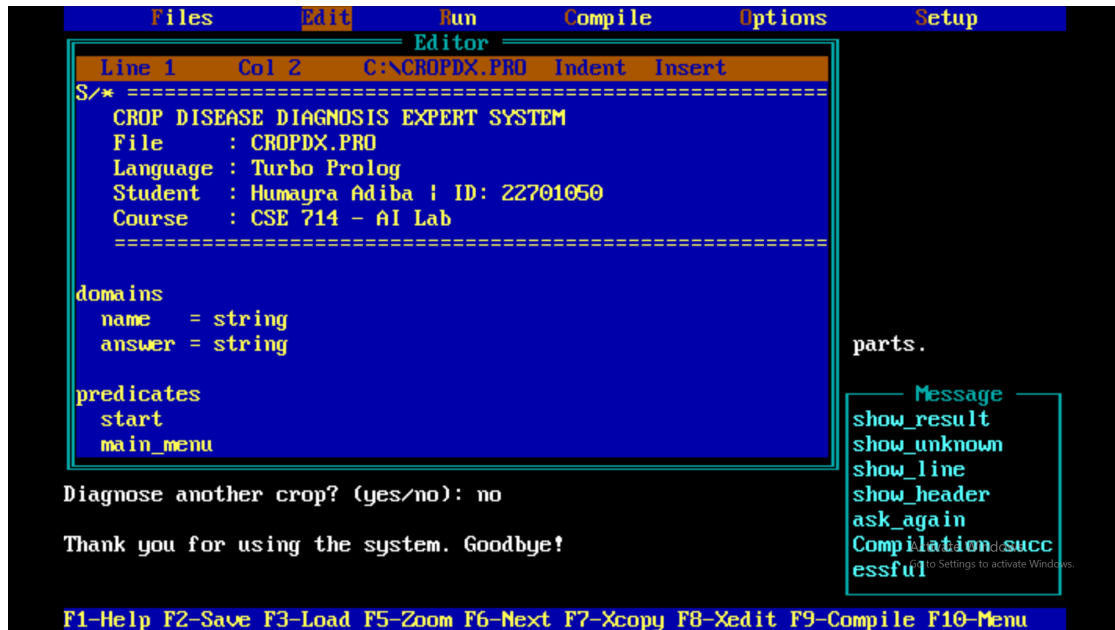


Figure 1: Turbo Prolog IDE showing CROPDX.PRO loaded and compiled successfully inside DOSBox.

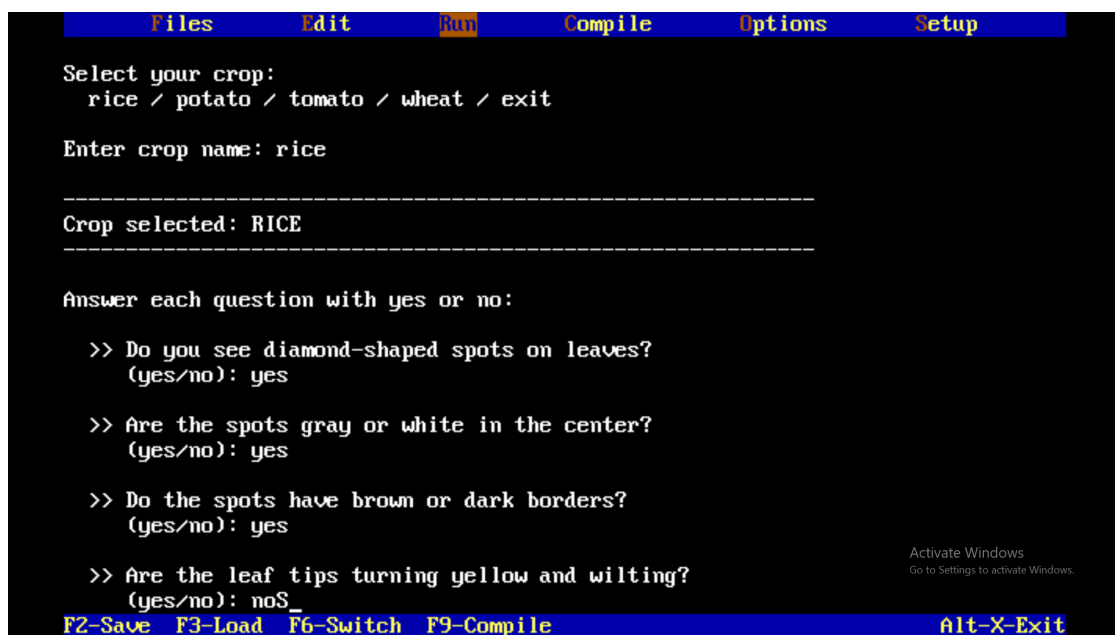


Figure 2: User interaction: crop selection and symptom questionnaire for Rice Blast diagnosis.

```

Files      Edit      Run      Compile      Options      Setup

>> Do you see water-soaked streaks on leaf margins?
    (yes/no): no

>> Are leaves drying out from the tips downward?
    (yes/no): no

-----
DIAGNOSIS RESULT
-----

DISEASE   : RICE BLAST
CAUSE     : Caused by fungus Magnaporthe oryzae.
TREATMENT : Apply Tricyclazole or Carbendazim. Remove infected parts.
PREVENTION: Use resistant varieties. Avoid excess nitrogen.

-----
Diagnose another crop? (yes/no): S_
F2-Save  F3-Load  F6-Switch  F9-Compile  Alt-X-Exit
Activate Windows
Go to Settings to activate Windows.

```

Figure 3: Diagnosis output screen showing Disease, Cause, Treatment, and Prevention for Rice Blast.

11.2 Sample Session Transcript

Sample Session: Rice Blast Diagnosis

```

=====
CROP DISEASE DIAGNOSIS EXPERT SYSTEM
AI Lab Project - Turbo Prolog
=====

```

Select your crop:

rice / potato / tomato / wheat / exit

Enter crop name: rice

Crop selected: RICE

Answer each question with yes or no:

```

>> Do you see diamond-shaped spots on leaves?
    (yes/no): yes

```

>> Are the spots gray or white in the center?

(yes/no): yes

>> Do the spots have brown or dark borders?

(yes/no): yes

>> Are the leaf tips turning yellow and wilting?

(yes/no): no

>> Do you see water-soaked streaks on leaf margins?

(yes/no): no

>> Are leaves drying out from the tips downward?

(yes/no): no

DIAGNOSIS RESULT

DISEASE : RICE BLAST

CAUSE : Caused by the fungus Magnaporthe oryzae.

TREATMENT : Apply Tricyclazole or Carbendazim.

Remove infected parts.

PREVENTION: Use resistant varieties.

Avoid excess nitrogen. Proper spacing.

Diagnose another crop? (yes/no): no

Thank you for using the system. Goodbye!

12 Advantages

1. **Offline operation:** No internet required — deployable on any DOS-capable hardware.
2. **Transparent reasoning:** Rules are readable English-like Prolog clauses — auditable by domain experts.
3. **Extensible:** New crops and diseases can be added by appending clauses without modifying the inference engine.
4. **Consistent advice:** Unlike human experts, the system provides uniform, unbiased recommendations every time.
5. **Low cost:** Requires only DOSBox (free) and a standard PC — no cloud subscription needed.
6. **Educational value:** Demonstrates core AI concepts: knowledge representation, rule-based reasoning, backtracking.
7. **Session-aware:** Multiple diagnoses per run without restarting the program.

13 Limitations

Known Limitations

1. **Binary symptom only:** Only yes/no answers accepted; no provision for severity levels or numeric measurements.
2. **Limited disease coverage:** Only 2 diseases per crop; many real diseases remain unhandled.
3. **No image input:** Cannot process photographs or sensor data.
4. **No certainty factors:** No probability weighting — a partial symptom match does not trigger a diagnosis.
5. **CLI only:** Text-based interface may be challenging for illiterate or semi-literate users.
6. **Static knowledge base:** Cannot learn from new cases; manual updating is required.
7. **No multi-disease support:** If a plant has two simultaneous diseases, only one can be diagnosed per session.

14 Future Improvements

1. **Expand knowledge base:** Cover 20+ diseases per crop using community-sourced phytopathology data.
2. **Certainty factors:** Assign confidence weights to rules (e.g. Mycin-style CF values) to handle partial symptom matches.
3. **Graphical interface:** Port to a modern Prolog (SWI-Prolog + XPCE) or develop a web front-end.
4. **Image-based input:** Integrate a CNN leaf-disease classifier (e.g. PlantVillage model[10]) to replace manual symptom entry.
5. **Multilingual support:** Translate symptom questions into Bengali, Hindi, and other regional languages.
6. **Mobile application:** Android app with offline SQLite knowledge base for field use.
7. **Machine learning integration:** Hybrid approach combining rule-based diagnosis with ML-trained classifiers.
8. **Case history logging:** Store diagnosis sessions to build a local epidemiological database.

15 Conclusion

This project successfully demonstrates the design, implementation, and testing of a rule-based expert system for crop disease diagnosis using Turbo Prolog. The system correctly diagnoses eight diseases across four staple crops (Rice, Potato, Tomato, Wheat) through a structured symptom questionnaire, providing actionable treatment and prevention advice.

The implementation validates core AI principles:

- **Knowledge engineering** — extracting and formalising expert knowledge.
- **Logic-based reasoning** — Prolog's backtracking as the inference engine.
- **Pattern matching** — clause unification as the diagnostic mechanism.
- **Graceful degradation** — the wildcard fallback for unknown conditions.

While the current system is intentionally scoped for educational purposes, its architecture is sound and extensible. The project underscores that classical AI paradigms — expert

systems, logic programming — retain significant practical value[9], particularly for offline, resource-constrained agricultural advisory applications.

Final Verdict

CROPDX achieves all stated objectives: it is functional, extensible, well-documented, and demonstrates the power of declarative AI programming for real-world problem solving.

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